

Feature Scaling, Model Optimization & Performance Comparison

Artificial Intelligence & Machine Learning Internship Task 2 Report

1. Methodology

The objective of this task was to compare multiple regression models for predicting house prices using the California Housing dataset. The implementation was completed in Python using Pandas, NumPy, Matplotlib, and Scikit-learn.

The California Housing dataset was loaded using Scikit-learn's `fetch_california_housing()`. The eight numerical input features were separated from the target variable, named **HousePrice** in the notebook. The data was divided into training and testing sets using an 80:20 split with `random_state=42`.

Feature scaling was then performed using **StandardScaler**. The scaler was fitted only on the training data and then used to transform both the training and test sets. This prevents information from the test set from being used during preprocessing.

Three regression models were trained and evaluated: **Linear Regression**, **Ridge Regression** with $\alpha = 1.0$, and a **Decision Tree Regressor** with maximum depth 5. Model performance was compared using Root Mean Squared Error (RMSE) and R^2 score.

2. Model Comparison and Results

The model comparison produced the following results on the test dataset:

Model	RMSE	R^2 Score
Linear Regression	0.746	0.576
Ridge Regression	0.746	0.576
Decision Tree	0.724	0.600

The Decision Tree Regressor achieved the lowest RMSE of **0.724234** and the highest R^2 score of **0.599732** among the three evaluated models. Therefore, the notebook selected the Decision Tree as the final model using the lowest-RMSE criterion.

Linear Regression and Ridge Regression produced very similar results. Linear Regression achieved an RMSE of 0.745581 and R^2 of 0.575788, while Ridge Regression achieved an RMSE of 0.745557 and R^2 of 0.575816. The small difference indicates that the selected Ridge regularization setting produced only a minor change relative to ordinary Linear Regression on this test split.

3. Visual Validation

After selecting the Decision Tree, the model was retrained on the scaled training data and used to generate predictions for the test data. An Actual vs Predicted scatter plot was created to visually compare the predicted house prices with the actual values. A diagonal reference line was included to represent ideal predictions.

The final notebook reports an RMSE of **0.724** and an R^2 score of **0.600** for the selected Decision Tree model. The trained model was also saved as `best_california_housing_model.pkl` using Joblib.

4. Conclusion

This task demonstrated a complete model comparison workflow for a regression problem. The California Housing dataset was prepared using an 80:20 train-test split and StandardScaler, followed by training three different regression algorithms. RMSE and R^2 were used to compare their performance.

Based on the test-set comparison, the Decision Tree Regressor was selected as the final model because it produced the lowest RMSE (0.724) and the highest R^2 score (0.600) among the evaluated models. The project also demonstrated visual validation through an Actual vs Predicted plot and saved the selected model for possible future use.

5. Possible Improvements

- Experiment with different Decision Tree depths and other tree parameters.
- Evaluate additional regression algorithms such as Random Forest or Gradient Boosting.
- Use cross-validation to obtain a more robust estimate of model performance.
- Perform systematic hyperparameter tuning instead of testing only one Ridge alpha and one tree depth.